



Shenzhen Hi-Link Electronic Co., Ltd

ZW3021

Fingerprint module manual

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1. Overview

ZW3021 is a capacitive fingerprint sensor MCU, A multifunctional illuminated fingerprint module that integrates fingerprint algorithm. The MCU is embedded with its own dual-mode algorithm, with good FRR/FAR performance and fast recognition speed.

The fingerprint module adopts low power consumption, small size design, high static electricity level, fast response, and excellent image quality. It adopts a press type input scheme, and can be unlocked with less input times by pressing once.

This module uses Command command communication for external communication and supports UART interface communication, making it convenient for users to integrate quickly.

2. Characteristics

- Press type fingerprint recognition
- Passive fingerprint detection
- Output 8-bit 256 grayscale 3D image
- 360°fingerprint input and matching
- The resolution is 508DPI
- Ultra low power design (6UA@7HZ)
- ESD: 15KV for air/8KV for contact with soil

3. Typical applications

Lock

4. Fingerprint detection area features

Table 1 Characteristics of Fingerprint Detection Area

Parameter	Numerical value	Unit
resolution	508	DPI

5. Performance parameters

Table 2 Performance Parameters Table

Serial number	Project	Parameter
1	Compare time	1:1<450ms
2	Startup time	<130ms
3	Feature learning and storage time	<100ms
4	interface protocol	RS232 (TTL)
5	Baud rate	57600bps
6	Comparison method	1:1&1: N
7	Rejection rate (FRR)	<5%
8	False Recognition Rate (FAR)	<0.001%
9	storage capacity	Comes standard with 100 pieces
10	demand times	1000000 times

6. DC parameters

Table 3 DC Parameters

Parameter	Description	Minimum	Typical value	Maximum	Unit
power supply voltage	VCC-S, Fingerprint sensor power supply	3.15	3.3	3.6	V
	VCC-D: DSP power supply	3.0	3.3	3.6	V
power supply current	Working mode: VCC-S and VCC-D are both powered on, and the module is in the mapping algorithm operation mode	/	45	/	mA
	Sleep mode: VCC-D loses power, DSP loses power, fingerprint sensor is in finger detection mode	/	6	/	μA

7. Limit parameters

Table 4 Limit Parameters

Parameter	Mark	Minimum	Maximum	Unit
Storage temperature	Tstg	-40	85	°C
Operating Temperature	Totg	-20	75	°C
power supply voltage	-	-0.5	3.6	V

8. Module Pin Definition

Table 5 Definition of ZW3021 module pins

NO.	Name	Description
1	VCC-S	Fingerprint sensor sensor power input terminal+3.3V
2	INT	Fingerprint sensor touch sensing signal output, default high level is valid.
3	VCC-D	Module DSP power supply positive input terminal+3.3V
4	UART-TX	Serial data output, TTL logic level.
5	UART-RX	Serial data input, TTL logic level.
6	GND	signal ground

Note:

VCC-S: Fingerprint sensor power input, requires constant power supply, input voltage 3.3V.

INT: The Interrupt pin of the fingerprint sensor changes its voltage level from low to high when pressed by a finger, and remains low for other times.

VCC-D: DSP power input, powered on when pressed with fingers; When there is no finger pressing, power off; To achieve low power consumption, the input voltage is 3.3V.

UART_TX: Module MCU serial port sending pin, connected to terminal MCU UART RX pin, 3.3VTTL level, default baud rate 57600, 8N2, can send control commands to the module

through UART interface.

UART_RX: Module MCU serial port receiving pin, connected to terminal MCU UARTTX pin, 3.3VTTL level. GND: Ground signal.

9. Module outline drawing



Figure 1 Outline drawing of ZW3021 module

10. Module Three Views

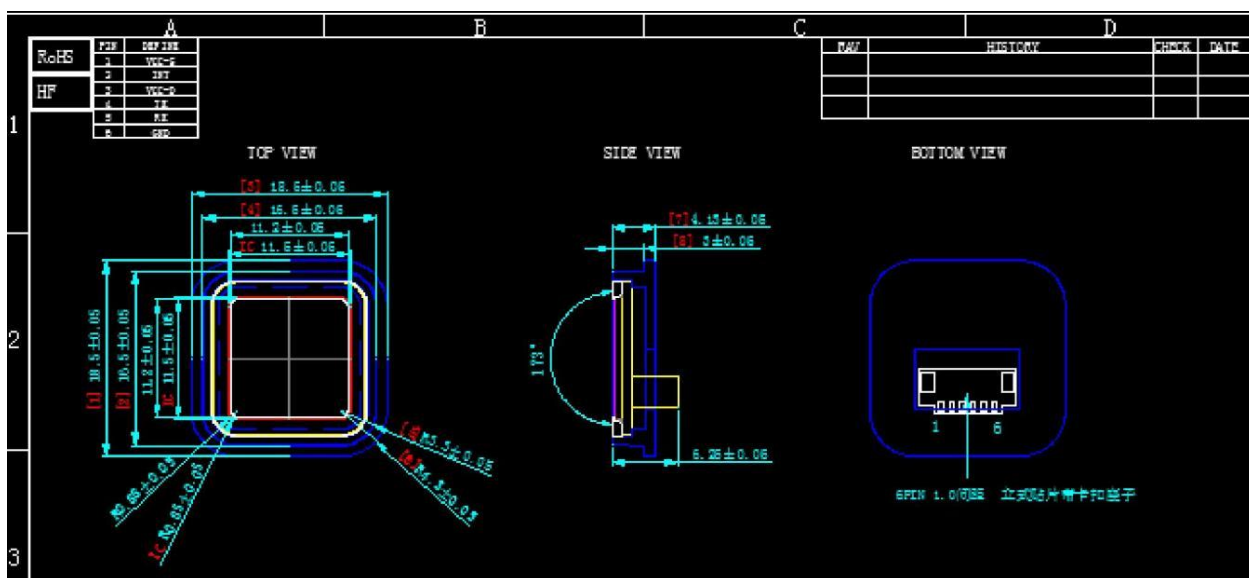


Figure 2 ZW3021 module size diagram

11. Module reliability requirements

Serial number	Test Project	Test conditions	Sample size	Allowable number of failures	Judgment criteria
1	Alternating damp heat	The temperature of the box is set to 25 °C and the humidity is 95%. It takes 3 hours to rise to 55 °C while keeping the humidity constant for 9 hours. Then, it takes another 3 hours to drop from 55 °C to 25 °C while keeping the humidity constant for 9 hours. This is counted as one cycle and is repeated three times for a total of 72 hours.	6pcs/batch, 1 batch	0	After the experiment, the appearance of the sample was normal and the functional performance test was qualified.
2	temperature cycling	-40 °C/15min~85 °C/15min, conversion time<5 minutes, 96 cycles.	12pcs/batch, 1 batch	0	
3	Damp heat bias	85 °C, 85% RH, normal operation for 96 hours.	12pcs/batch, 1 batch	0	
4	High temperature work	85 °C, normal operation for 96 hours.	12pcs/batch, 1 batch	0	
5	Low temperature operation	-40 °C, normal operation for 96 hours.	12pcs/batch, 1 batch	0	
6	high temperature storage	85 °C, 96 hours.	6pcs/batch, 1 batch	0	
7	Low-temperature storage	-40 °C, 96 hours.	6pcs/batch, 1 batch	0	
8	mechanical shock	Half sine pulse 30g, 18ms, 3 times in each direction ($\pm x$, $\pm y$, $\pm z$).	3pcs/batch, 1 batch	0	
9	mechanical vibration	5Hz~20Hz 0.96 (m/s^2) 2/Hz; 20Hz~500Hz, -3dB/octave band.		0	
10	pencil hardness test	Using Mitsubishi pencils, test the coating surface of the sample according to the test method in the reference standard, with a load of 750g, an angle of 45 ° between the pencil and the horizontal plane, sliding at least 7mm, and making one line at each of the five different positions.	3pcs/batch, 1 batch	0	There are no permanent indentations on the surface (small indentations are allowed at the starting point of 0-2mm), and visible scratches or abrasions are not allowed on the surface.
11	Artificial sweat	Soak the sample surface in artificial	3pcs/batch, 1	0	The surface coating

	resistance test	sweat (acidic formula, alkaline formula) and gently wipe it with a dust-free cloth for 2 minutes, then leave it at room temperature for 2 hours	batch		has no obvious discoloration, blistering, peeling, cracking or other abnormal phenomena.
12	Durable cosmetics testing	Apply Vaseline special moisturizing lotion onto the surface of the coating, Perform 2 cycles (48 hours in total) according to the following steps: The temperature of the box is set to 25 °C and the humidity is 95%. It takes 3 hours to rise to 55 °C while keeping the humidity constant for 9 hours. Then, it takes another 3 hours to drop from 55 °C to 25 °C while keeping the humidity constant for 9 hours, which is counted as one cycle. 25 °C, constant humidity, held for 9 hours, counted as 1 cycle	3pcs/batch, 1 batch	0	After the experiment, there were no obvious discoloration, bubbles, cracks, coating peeling or other abnormalities on the surface of the sample, and the adhesion test reached 4B.
13	Coating adhesion test	Use a hundred grid test knife to draw 10 x 10 1mm x 1mm small grids on the surface of the sample coating, and each line should be deep enough to reach the bottom layer of the coating; Use a brush to clean the fragments in the testing area; Use 3M 610 # adhesive tape to firmly adhere to the small grid being tested, and use an eraser to forcefully squeeze the tape to remove any air bubbles between the tape and the coating, in order to increase the contact area and strength between the tape and the tested area; Grasp one end of the tape with your hand, pull off the adhesive paper in the direction of 60 ° within 0.5s~1s, and test once.	3pcs/batch, 1 batch	0	After the experiment, it meets the requirement of 4B.
14	Boiling test	Boil the sample in water at 80 °C± 2 °C for 30 minutes, and leave it at room temperature for 2 hours before conducting an adhesion test.	3pcs/batch, 1 batch	0	The surface coating has no obvious discoloration, peeling, cracking or other phenomena; The adhesion test reached 3B.
15	Alcohol Friction Test	Dip pure cotton cloth in alcohol with a concentration of ≥ 99.5%, wrap it on a special 500g lifting platform, and rub the sample surface 250 times	3pcs/batch, 1 batch	0	The surface coating has no obvious scratches or peeling phenomenon.

		at a speed of 40 times/minute to 50 times/minute.			
16	RCA abrasion resistance test	Using a wear tester and specialized paper tape, with a continuous friction load of 175g. UV: 200 cycles; PU: 150 laps.	3pcs/batch, 1 batch	0	Sample surface without exposed substrate
17	Electrostatic Discharge Immunity Test ESD (HBM)	Determine according to product design requirements.	3pcs/batch, 1 batch	0	The sample performance test passed after the experiment.
18	Latch Effect Test (LU)	Determine according to product design requirements.	3pcs/batch, 1 batch	0	The sample performance test passed after the experiment.

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